AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM276
Software Development Processes and Methodologies

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 17 /Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

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Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 3 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which essential attribute of good software states "software should not cause physical or economic damage in the event of system failure":
   a. Maintainability b. Dependability and Safety c. Efficiency d. Acceptability

2. Which software process model is purely plan-driven:
   a. Waterfall b. Incremental development c. Integration and configuration d. V-model

3. To which non-functional requirements category does "Efficiency Requirements" belong:
   a. Product requirements b. Organizational requirements c. External requirements d. Hardware requirements

Part II – Short Answer Questions — 12 Marks

*Answer **all** the following short answer questions. Each question is worth 4 Marks.*

1. List with a brief explanation 2 of the main design activities

2. Discuss the below Extreme Programing practices
   a. Test-first development
   b. Pair programming

3. Given the below User Requirement definition, provide 2 System Requirements Specification.
   The iLearn system shall generate weekly reports showing the number of registrations per course and the pass rate.

Part III – Problem Solving Questions — 15 Marks

*Answer **all** the following problem-solving questions. Each question is worth 7.5 Marks.*

1. Develop an activity diagram for the process of reserving a train ticket and make sure to cover the following:
   a. Must include a minimum of 6 activities
   b. Must have at least one decision and one merge node
   c. Must have at least 2 roles represented by swimlanes
   d. Must use the synchronization bars (Fork and Join) at least once.

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2. Develop a Use Case diagram for an LMS with the following requirements:
   a. The main actor is a user that can only read contents online, download files, and submit assignments.
   b. A tutor is a user that can create discussion topics, and download assignments
   c. A coordinator is a tutor that can edit content, and upload files.

   Important: Make sure to use Generalization properly and use stereotypes at least once.

End of questions

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